

MAE 143 A - Fall 2019

Maurício de Oliveira

Homework 1

1. Exercise 1.6 from Boulet FSS
2. Exercise 1.8 from Boulet FSS
3. Exercise 1.11 from Boulet FSS
4. Let

$$f(t) = \begin{cases} 0, & t < 0, \\ 1, & 0 \leq t < 1, \\ e^{-t+1}, & t \geq 1 \end{cases}$$

and answer the following questions:

- a. Sketch the signal $f(t)$;
- b. Sketch the signals
 1. $g_1(t) = f(-t)$
 2. $g_2(t) = -f(t - 1)$
 3. $g_3(t) = 2f(2t)$
 4. $g_4[n] = f(0.5n)$, $n \in \mathbb{Z}$
 5. $g_5(t) = \int_{-\infty}^t f(\tau) d\tau$
 6. $g_6(t) = \dot{f}(t)$
- c. Calculate:
 1. $\int_{-\infty}^{\infty} f(t)$
 2. $\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T f(t)$
5. Let $f(t)$ from question 4 be the impulse response of a time-invariant linear system and answer the following questions:
 - a. Is the system causal?
 - b. Is the system memoryless?
 - c. Use the convolution formula

$$y(t) = \int_{-\infty}^{\infty} f(t - \tau) u(\tau) d\tau$$

to calculate the response of the system to a unit step input.

- d. Is the response you calculated in c. bounded?
6. When working with time series, a commonly used filter is a *moving average* filter, as in

$$y[k] = \frac{1}{N} \sum_{i=0}^{N-1} u[k - i].$$

Such a filter is a linear time-invariant filter. Calculate and sketch its impulse response and interpret the above formula in terms of a convolution.